

REVERSIBLE MASKING: A NOVEL FAULT-DIAGNOSEDPriyanki Roy¹ and Samik Banerjee¹¹CSE Department, Heritage Institute of Technology, Kolkata, WBUT, West Bengal, India
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Abstract: This paper suggests a novel design of reversible masking circuit for Quantum Cryptography. Quantum computation uses quantum properties to represent data and perform reversible operations on data. In this paper, we proposed to design a reversible masking logic and implement the masking logic using basic quantum gates. The masking expression is thus transformed into a Positive Polarity Reed Muller Expression to calculate its nonlinearity. We also proposed a novel quantum gate design, namely M-gate, based on the basic quantum gates. To analyze the design quantum cost is calculated can improve the circuit comparatively another masking circuit. From this circuit we are getting high nonlinearity which is much better than another masking circuit. Here also we test the circuit by using single Missing Gate Fault Model and Multiple Missing Gate Fault Model.

Keywords: Masking, reversible, quantum cryptography, nonlinearity, boolean logic.

INTRODUCTION

The researchers are trying so hard to develop a practical quantum computer because atoms change their energy states very quickly than even the fastest computer processors. But, for the novel methods of information processing and transmission Quantum information theory follow the laws of quantum mechanics that would be impossible or very hard to achieve classically [1]. Day by day, technology is trying to contain great computing power in smaller and smaller devices [2]. For designing a general-purpose quantum computer, classical boolean logic cannot implemented by quantum devices. A straightforward realization of quantum boolean logic is to use auxiliary qubits as intermediate storage. However, to build a general-purpose computer, it is very necessary use the type of nanoscale quantum devices which has the ability to implement classical boolean logic. If large-scale quantum computers can be built, they will be able to solve certain problems much faster than any current classical computers.

We can simulate a classical logic circuit using a quantum circuit. But some classical circuits cannot be simulated directly by quantum circuits because unitary quantum logic gates are inherently reversible. But also several classical logic gates such as NAND gates are inherently reversible. A classical circuit can be replaced by an equivalent circuit, which contains reversible logic gates [3]. Actually the main component of quantum logic gate is reversible logic gate. 'Toffoli gate' is a reversible gate. Thus Toffoli gate is a reversible gate which also has an inverse. Toffoli gate can be used to simulate NAND gate. Using an equivalent reversible circuit, any classical circuit can be simulated. Several masking techniques used today has a direct implementation in the classical world. The researchers are making several attempts to implement the masking techniques in the quantum scenario through a duplication of the environment in the reversible logic so that there should be high ratio of error-detection much faster providing higher throughput for testing. This paper suggests such a scenario of a masking expression. The paper is organized in the way to understand the flow from formulization of a mask expression to its eventual quantum implementation. The formulation is discussed in later with its implementation in succeeding parts. Since masking is a major tool in the implementation of security of the circuit, the quantum circuit is analyzed and discussed in the Section IV. The circuit is thus analyzed for its robustness on the basis of gate-count and Quantum. The circuit is also tested for faults with the generation of Test Vectors.. The experimental results are hence shown.

REVERSIBLE MASKING

Masking is the one of the technique to secure cryptographic circuit and it makes each probable attacked signal unpredictable. In this approach, each of probable attacked signal b is represented by

$$b_m = b \oplus m_b,$$

Where, m_b is a uniformly distributed random variable
i.e. $P(m_b = 0) = P(m_b = 1) = 1/2$.

and is independent of b .

The b_m also is a uniformly distributed random variable. In the masking-approach protected masked cell [4] can be replaced each cell, which is achieved by exclusive-or (XOR).

For e.g. A 2-input XOR function $g = a \oplus b$ can be replaced by

$$g_m = a_m \oplus b_m$$

$$\text{and } m_g = m_a \oplus m_b.$$

The m_g is called its correction mask, where

$$b_m = b \oplus m_b$$

$$\text{and } a_m = a \oplus m_a.$$

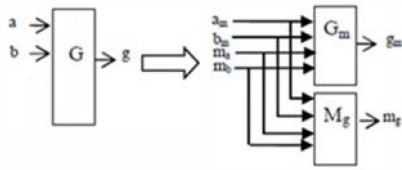


Fig.1 A 2-input unmasked cell G and a corresponding masked cell G_m and correction mask generator M_g

We proposed one masked expression using quantum logic for two-input classical AND gate $g=a.b$, which is derived as follows:

$$g_m = m_g m_a m_b + b_m m_a' a_m' + m_a b_m' m_g + a_m b_m' m_b'$$

$$\text{where, } g_m = a_m \oplus b_m,$$

$$m_g = m_a \oplus m_b$$

$$b_m = b \oplus m_b,$$

$$a_m = a \oplus m_a,$$

$$m_a = a'$$

- The g_m represents mask generator.
- The m_g represents the correction mask.
- This expression also reversible for suitable m_b .

The truth table of this proposed mask is:

a	b	g_m	m_g
1	1	0	0
0	0	1	1
0	1	0	1
1	0	1	0

We can minimize the mask by using Karnaugh map.

$$\text{Min. of } g_m = b'm_b' + am_b.$$

$$\text{Min. of } m_g = am_b' + bm_b.$$

$$x' = 1 \oplus x.$$

Reed Muller Form

In order to improve the security of masking circuit we can transform the masking circuit to Reed Muller form

[5]. Reed Muller (RM) circuits are easier to test and require fewer logic gates than traditional sum-of-product forms generally. In this paper we represent a Positive Polarity Reed-Muller expression (PPRM). Reed Muller logic style can save much area than AND-OR cryptographic application. So we convert the AND-OR mask is to Reed Muller form (AND-XOR). We further transform it to Positive Polarity Reed Muller expression.

Positive Polarity Reed Muller (PPRM)

Positive Polarity Reed Muller expression contains as one of its prime implicants

$$\text{e.g. } 1 \oplus c \oplus ab \oplus ac$$

We can derive the masking expression to Reed-Muller form such that

$$g_m = 1 \oplus b \oplus m_b \oplus bm_b \oplus am_b.$$

This is Positive Polarity Reed Muller expression which is represented by following circuit (fig. 2). We can derive the boolean expression of correction mask (m_g) to Reed-Muller form, which is represented by following circuit (fig. 3) such that

$$m_g = a \oplus am_b \oplus bm_b$$

This is a correction mask REED MULLER expression. From any PPRM expression we can easily calculate the Non-Linearity of the expression.

REVERSIBLE CIRCUIT IMPLEMENTATION

Here we implement two reversible circuits for g_m and m_g .

$$g_m = 1 \oplus b \oplus m_b \oplus bm_b \oplus am_b$$

For g_m , we design M gate which has four levels; A Level, B Level, C Level, D Level.

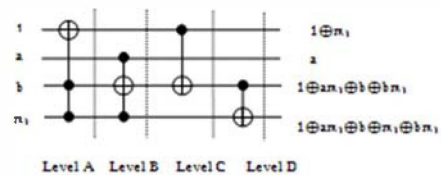


Fig. 2 Gate Level Realization of g_m

$$m_g = a \oplus am_b \oplus bm_b$$

For m_g , we design C gate which has three levels; P Level, Q Level, R Level.

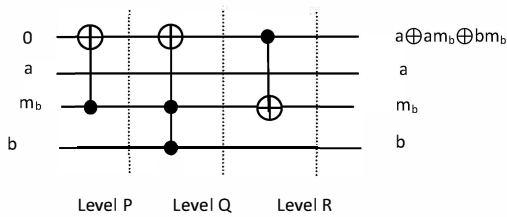


Fig. 3 Gate Level Realization of m_g

Combination of g_m and m_g

In this paper we thus combine g_m and m_g gate. To represent this compound circuit (CM) we are using one extra input line '0' i.e. constant for every time.

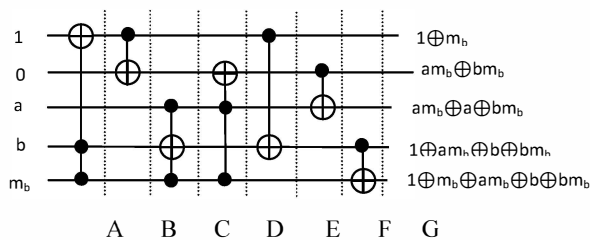


Fig. 4 Compound Gate

Gate Matrix for g_m

The gate Matrix for g_m is represented as follows.



Gate Matrix for m_g

The gate Matrix for m_g is represented as follows.



Gate Matrix for Compound Gate (CM)

The gate Matrix for M is represented in Yale Format Sparse Matrix as follows:

$$A = [1]$$

$$\text{IA} = [k : k \in \mathbb{N}, 0 \leq k \leq 31]$$

$$JA = \begin{bmatrix} 0 & 1 & 3 & 29 & 4 & 10 & 7 & 22 & 12 & 13 & 15 & 17 & 8 & 6 & 11 & 26 & 19 & 18 \\ 16 & 14 & 23 & 25 & 20 & 5 & 31 & 30 & 28 & 2 & 27 & 21 & 24 & 9 \end{bmatrix}$$

NON LINEARITY

The Hamming weight of a binary string T , denoted by $wt(T)$, is the number of 1's in T . An n -variable function f is said to be balanced if its truth table contains equal no. of 0's and 1's, i.e. $wt(f)=2^{n-1}$. The Hamming distance between two equidimensional binary strings T_1 and T_2 denoted as $dist(T_1, T_2)$ is the no. places where T_1 and T_2 do not match.

i.e. $\text{DIST}(T_1, T_2) = \text{WT}(T_1 \oplus T_2)$.

Function of degree at most one are called affine functions (A(n)). An affine function with constant term equal to zero is called a linear function (L(n)). The set of all n-variable affine function is denoted by A(n) (respectively L(n)). Therefore, for

A	B	m _b	g _m	m _g	g ₁	g ₂	g ₃	g ₄	P	Q	R	S
0	0	1	0	1	1	0	0	0	0	1	1	1
0	0	1	0	1	1	0	0	0	0	1	1	1
1	0	0	0	1	0	0	0	0	1	1	1	0
1	1	0	0	1	0	0	0	0	1	1	1	0
0	0	1	1	0	0	0	1	0	1	0	1	0
0	1	1	1	1	0	0	0	0	1	1	1	0
1	0	0	1	1	1	0	0	0	0	1	1	1
1	1	0	1	1	1	0	0	0	0	1	1	1
0	0	0	0	1	0	0	0	0	1	1	1	0
0	1	0	0	0	0	0	1	0	1	0	1	0
1	0	1	0	1	1	0	0	0	0	1	1	1
1	1	1	0	1	1	0	0	0	0	1	1	1
0	0	0	1	1	1	0	0	0	0	1	1	1
0	1	0	1	0	1	0	1	0	0	0	1	1
1	0	1	1	0	0	0	1	0	1	0	1	0
1	1	1	1	0	0	0	1	0	1	0	1	0

Table 1: Non Linearity Calculation

fixed $w=(w_1, \dots, w_n) \in \{0,1\}^n$, an affine functions of the form

$$w \cdot x \oplus c,$$

Where $c \in \{0,1\}$ is a constant

$$\text{And } w \cdot x = w_1 x_1 \oplus \dots \oplus w_n x_n.$$

For example, take $(0,1,0,1) \in \{0,1\}^4$ and $c = 1$, then the corresponding affine function is $x_2 \oplus x_4 \oplus 1$.

The non linearity of n -variable function f is defined as

$$nl(f) = \min_{g \in \mathcal{A}(n)} (\text{dist}(f, g)).$$

i.e. the min. dist. from the set of all n -variable affine functions.

For function $f \in \beta_n$, a nonzero boolean function $g \in \beta_n$ is called annihilator of f if

$$f(x)g(x) = 0, \text{ for all } x \in \{0,1\}^n.$$

Let's define the following set,

$$AN(f) = \{g \in \beta_n \mid g \text{ nonzero } f \cdot g = 0\},$$

i.e. set of all annihilator of f .

On calculating nonlinearity of n -variable function we get

$$nl(g_m) = 8, \text{ and } nl(m_g) = 8.$$

Where, P is denoted by $g_m \oplus g_1$,

Q is denoted by $g_m \oplus g_2$,

R is denoted by $m_g \oplus g_3$,

and S is denoted by $m_g \oplus g_4$.

GATE COUNT

We convert the quantum circuit to classical circuit by using Xilinx software. Converting to classical circuit we get the gate count of the M Gate is 11, C gate is 8, CM is 14. M gate consist of 8 Classical AND gate and 3 OR gate. C gate consist of 6 Classical AND gate and 2 OR gate. M gate consist of 10 Classical AND gate and 4 OR gate.

QUANTUM COST

Quantum cost of the proposed Model (g_m) is 12. The Quantum Cost (QC) of a reversible circuit can be measure as the number of 1×1 or 2×2 reversible quantum or logic gates which are needed to realize the circuit. Here this proposed gate consists of two Toffoli gates and two C-NOT gates. As proposed previously [6], the quantum cost of the Toffoli gate is 5 and quantum cost of C-NOT gate is 1.

FAULT TESTING

By reviewing this circuit we find the fault of this circuit. In this paper without using stuck-at fault model [8] we are using two types of testing i.e. Single Missing Gate Fault (SMGF) and Multiple Missing Gate Fault (MMGF). This testing requires minimal set of test

vector to cover all the faults. One test vectors are needed to detect SMGF i.e. 1101, 1111, 1100. MMGF is also two types i.e. 2-MMGF and 3-MMGF. To detect 3-MMGF 1111 test vector, and to detect 2-MMGF 1101 test vector are required.

EXPERIMENTAL RESULTS

Gates	Non-Linearity	Gate Count in Classical circuit
Conventional Gate	7	14
M Gate	11	11
C Gate	8	8
CM Gate	8	14

Table 2: Comparison Chart 1.

Reversible circuits	Quantum Cost	Gate Count
M Gate	12	4
C Gate	11	3
CM Gate	19	7
hwb4-11-21	21	11
hwb4-11-23	23	11
hwb4_fc	81	11
hwb4tc	63	17
mspk_4_49_12	30	12
mspk_4_49_13	29	13
mspk_4_49_14	28	14
mspk_4b15g_1	39	15
mspk_4b15g_2	31	15
mspk_4b15g_3	33	15
mspk_4b15g_4	35	15
mspk_4b15g_5	29	15
mspk_hwb4_12	20	12

Table 3: Comparison Chart 2.**CONCLUSIONS**

The proposed design gives a high nonlinearity coefficient and being reversible can be used in high performance computing domain for a very high security measure. this also gives the low quantum cost and it requires minimum number of gate than another conventional circuit. the novel idea of quantum masking technique introduced in this paper will throw light on this domain to encourage researchers for a new era of quantum computation.

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REFERENCES

- [1] JaromirFiurasek. (August 21, 2008): 'Linear optics quantum Toffoli and Fredkin gates',Dept of Optics, Palacky University.
- [2] Hidayath Ansari, AdityaParameswaran, LakulishAntani: 'Quantum cryptography and quantum computation', IIT, Bombay.
- [3] VlatkoVedral and Martin B. PlenioBlackett, (February 1, 2008): 'Basics of quantum computation', Laboratory, Imperial College, Prince Consort Road, London SW7 2BZ, U.K.
- [4] Kuan Jen Lin+, Shan Chien Fang, Shih Hsien Yang and Cheng Chia Lo: 'Overcoming glitches and dissipation timing skews in design of DPAResistant cryptographic hardware', Department of Electronic Engineering, Fu Jen Catholic University.
- [5] Jing Hu, GuangSheng Ma, Gang Feng: 'Efficient algorithm for positive-polarity reed-mullerexpansions of reversible circuits',Harcin University, Harbin.
- [6] H.R Bhagyalakshmi, M.K Venkatesha, (2010): 'An improved design of a multiplier using reversible logic gates', in International Journal of Engineering Science and Technology Vol. 2(8), 3838-3845.
- [7] J. P. Hayes, I. Polian, and B. Becker, (2004): 'Testing for missing gate faults in reversible circuits', in Asian Test Symp., pages 100–105.
- [8] K. C. Y. Mei, (July 1974): 'Bridging and Stuck-At Faults', in IEEE Transactions on Computers, Volume 23 Issue 7.